

The Cauchy–Kowalevski Theorem

Statement and Detailed Proof (Multi-Index Formulation)

MA PDE Lecture Notes

Multi-Index Notation

Let $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}_0^n$.

- $|\alpha| = \alpha_1 + \dots + \alpha_n$
- $\partial^\alpha = \partial_{x_1}^{\alpha_1} \dots \partial_{x_n}^{\alpha_n}$
- $\partial^0 u = u$

We write

$$\partial^k u = \{\partial^\alpha u : |\alpha| = k\}.$$

General Form of the PDE

Let $x = (x_1, \dots, x_n) \in \mathbb{R}^n$.

Consider an m th order PDE:

$$F(x, \{\partial^\alpha u : |\alpha| \leq m\}) = 0,$$

where F is real-analytic in all arguments near the origin.

Initial Surface and Data

Let the initial hypersurface be

$$\Gamma = \{x_n = 0\}.$$

Prescribe Cauchy data:

$$\partial^\beta u(x', 0) = g_\beta(x'), \quad |\beta| \leq m - 1,$$

where $x' = (x_1, \dots, x_{n-1})$.

Each g_β is real-analytic near $x' = 0$.

Non-Characteristic Condition

Assume the highest normal derivative appears with nonzero coefficient:

$$\frac{\partial F}{\partial(\partial_n^m u)}(0, \dots) \neq 0.$$

This is the non-characteristic condition for Γ .

It allows us to solve locally for $\partial_n^m u$.

Cauchy–Kowalevski Theorem

Theorem.

Suppose:

- F is real-analytic near the origin,
- the initial data g_β are real-analytic,
- the hypersurface $x_n = 0$ is non-characteristic.

Then there exists a neighborhood U of 0 and a unique real-analytic solution

$$u \in C^\omega(U)$$

satisfying the PDE and the prescribed Cauchy data.

Proof Strategy

- 1 Construct a formal power series in x_n .
- 2 Determine coefficients recursively.
- 3 Use analyticity and the non-characteristic condition.
- 4 Prove convergence via a majorant series.

Power Series Ansatz

Seek

$$u(x', x_n) = \sum_{k=0}^{\infty} \frac{x_n^k}{k!} h_k(x').$$

Then

$$\partial_n^k u(x', 0) = h_k(x').$$

For multi-index $\alpha = (\beta, k)$,

$$\partial^\alpha u(x', 0) = \partial_{x'}^\beta h_k(x').$$

Substitution into the PDE

Insert the series into

$$F(x, \{\partial^\alpha u\}).$$

Since F is analytic, we obtain

$$F = \sum_{j=0}^{\infty} \frac{x_n^j}{j!} R_j(x').$$

The PDE implies

$$R_j(x') = 0 \quad \forall j \geq 0.$$

Each R_j depends only on $h_0, \dots, h_j$.

Recursive Determination

For $j = 0, \dots, m - 1$:

$$h_j(x') = g_j(x').$$

For $j \geq m$:

$$R_j(x'; h_0, \dots, h_j) = 0.$$

The coefficient of h_j is

$$\frac{\partial F}{\partial(\partial_n^m u)} \neq 0.$$

Hence we solve for h_j uniquely.

Majorant Series Argument

Because all data are analytic, there exist convergent power series bounds.
Define

$$M_k = \sup_{|x'| \leq r} |h_k(x')|.$$

One proves inductively:

$$M_k \leq C k! R^{-k}.$$

Therefore

$$\sum_{k=0}^{\infty} \|h_k\| \frac{|x_n|^k}{k!}$$

converges in a polydisc.
Thus u is analytic.

Uniqueness

If two analytic solutions agree to order $m - 1$ on $x_n = 0$, then all normal derivatives agree inductively.

Analytic continuation implies

$$u \equiv v$$

in a neighborhood of the origin.

Conclusion

Under analyticity and the non-characteristic assumption:

- A local analytic solution exists.
- The solution is unique.
- The proof is constructive via power series.

This is the Cauchy–Kowalevski theorem.

Why Analyticity Matters

The Cauchy–Kowalevski theorem guarantees:

$$F(x, \{\partial^\alpha u\}) = 0$$

has a *local analytic solution* if

- F is real-analytic,
- the Cauchy data are real-analytic,
- the initial surface is non-characteristic.

Question: What if coefficients are merely C^∞ ?

Answer: Existence can fail completely.

The Lewy Non-Existence Example (1957)

In 1957, Hans Lewy constructed a smooth linear PDE

$$Lu = f$$

with smooth coefficients and smooth right-hand side for which *no local solution exists*.

Operator in $\mathbb{R}^3$:

$$L = \partial_x + i \partial_y + 2i(x + iy)\partial_t.$$

Lewy showed:

$$Lu = f$$

has no local C^1 solution near 0 for certain $f \in C^\infty$.

Structure of the Lewy Operator

Introduce complex variable

$$z = x + iy.$$

Then

$$L = \partial_x + i\partial_y + 2iz\partial_t = 2\partial_{\bar{z}} + 2iz\partial_t.$$

This is a first-order linear PDE with smooth coefficients.

The obstruction arises from non-integrability of the associated complex vector field.

Idea of Lewy's Argument

Suppose $Lu = f$ locally.

One studies compatibility conditions obtained by differentiating along commutators.

Compute:

$$[L, \bar{L}] = -4 \partial_t.$$

This generates a real direction transverse to the complex span of L and $\bar{L}$.

The system becomes overdetermined.

Certain smooth f violate the necessary compatibility conditions, yielding non-existence.

Why This Is Shocking

The equation:

- is linear,
- has C^∞ coefficients,
- is first order.

Yet:

No local solution exists.

This shows:

C^∞ regularity does not guarantee local solvability.

Lewy Operator: Commutator Geometry in Detail

Recall

$$L = \partial_x + i\partial_y + 2i(x + iy)\partial_t.$$

Introduce complex notation

$$z = x + iy, \quad \partial_{\bar{z}} = \frac{1}{2}(\partial_x + i\partial_y).$$

Then

$$L = 2\partial_{\bar{z}} + 2iz\partial_t.$$

Define also

$$\bar{L} = 2\partial_z - 2i\bar{z}\partial_t.$$

Computation of the Commutator

Compute the Lie bracket

$$[L, \bar{L}] = L\bar{L} - \bar{L}L.$$

Only the z -dependence of coefficients contributes.

Since

$$[\partial_{\bar{z}}, z] = 0, \quad [\partial_z, \bar{z}] = 0,$$

but

$$\partial_{\bar{z}}(\bar{z}) = 1, \quad \partial_z(z) = 1,$$

a direct calculation gives

$$[L, \bar{L}] = -4 \partial_t.$$

Geometric Meaning

At each point, the real and imaginary parts of L span a 2-dimensional distribution:

$$\mathcal{D} = \text{span}\{\Re L, \Im L\}.$$

Explicitly,

$$\Re L = \partial_x - 2y \partial_t, \quad \Im L = \partial_y + 2x \partial_t.$$

Their commutator gives

$$[\Re L, \Im L] = -4 \partial_t.$$

Thus the t -direction is generated only after taking a bracket.

Why This Causes Non-Solvability

If $Lu = f$ has a solution, then differentiating:

$$\bar{L}(Lu) = \bar{L}f.$$

Interchanging order gives

$$L(\bar{L}u) + [L, \bar{L}]u = \bar{L}f.$$

Since

$$[L, \bar{L}] = -4\partial_t,$$

we obtain the compatibility condition

$$-4\partial_t u = \bar{L}f - L(\bar{L}u).$$

For certain smooth f , this forces a contradiction.

Structural Interpretation

This bracket-generated transverse direction creates hidden compatibility conditions that smooth data need not satisfy.

Why Cauchy–Kowalevski Is Limited

The theorem requires:

analytic coefficients and analytic data.

But:

- Most PDEs arising in physics have only smooth coefficients.
- Analytic functions are rigid (no compact support).
- Small perturbations typically destroy analyticity.

Thus CK applies to a very narrow class of problems.

Cauchy–Kowalevski ignores:

- Hyperbolicity
- Ellipticity
- Energy structure
- Well-posedness in Sobolev spaces

It is purely an *analytic power series theorem*.

Modern PDE theory replaces analyticity with:

energy estimates and functional analysis.

Conceptual Takeaway

Cauchy–Kowalevski tells us:

Analytic PDE + analytic data $\Rightarrow$ analytic solution.

Lewy shows:

C^∞ coefficients $\nRightarrow$ local solvability.

Analyticity is a technical condition that guarantees a local analytic solution.

Contrast: Hyperbolic PDE Theory

Consider a first-order system

$$\partial_t u + \sum_{j=1}^n A^j(x, t) \partial_{x_j} u = F,$$

or the second-order wave equation

$$\partial_t^2 u - c^2 \Delta u = F.$$

These operators are **hyperbolic**, and

- Energy estimates

$$\|u(t)\|_{H^s} \leq C \left(\|u(0)\|_{H^s} + \int_0^t \|F(\tau)\|_{H^s} d\tau \right)$$

- Local (and often global) well-posedness
- Continuous dependence on data
- No analyticity assumption

CK vs. Hyperbolic Theory

	Cauchy–Kowalevski	Hyperbolic Theory
Coefficients	Analytic	C^k
Data	Analytic	H^s
Method	Power series	Energy estimates
Well-posedness	Not addressed	Central
Stability	Not addressed	Essential
Physical relevance	Limited	Fundamental

Key point:

CK proves existence in a rigid analytic category.

Hyperbolic theory explains why wave-type equations are stable and physically meaningful.